👩‍🏫 **Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 6 · STANDARD 6.NOS.1

Divide Whole Numbers by Fractions

Lesson 6-9 · Teacher Copy

Name: _____ **Date:** _____ **Class:** _____

My Notes

Level 3 Enrichment

🎯 **TODAY'S OBJECTIVES**

Content Objective: I can explain what dividing a whole number by a fraction means — how many pieces fit, or how much is in one whole — find the quotient with a model or by multiplying by the reciprocal, and check it by multiplying.

Language Objective: I can explain why multiplying by the reciprocal works, using the words fraction, reciprocal, quotient, and Keep, Change, Flip.

📖 See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.

✏️ **Fill in each blank as we go. Use the Word Bank to help you.**

📚 **WORD BANK — FILL EACH BLANK WITH THE BEST WORD**

Whole Number | **Fraction** | **Reciprocal** | **Keep, Change, Flip** | **Quotient**

Level 3: try the blanks from memory first, then check the bank.

1. A counting number like 0, 1, 2, or 3 with no fraction or decimal part is a _____.

2. A number that shows part of a whole, like 3/4, is a _____.

3 The fraction you get by flipping a fraction upside down is its _____ .

4 To divide by a fraction, I _____ : keep the first number, change ÷ to ×, and flip the second fraction.

5 The answer to a division problem is the _____ .

Write About the Math

How does dividing a whole number by a fraction help the detective?
¿Cómo ayuda dividir un número entero entre una fracción al detective?

Say your answer to a partner first. Then write it, using at least two of these words:

- ☐ **Whole Number** — *Número entero*
- ☐ **Fraction** — *Fracción*
- ☐ **Reciprocal** — *Recíproco*
- ☐ **Keep, Change, Flip** — *Mantener, cambiar, invertir*
- ☐ **Quotient** — *Cociente*

Model: *Each whole pound makes 2 half-pound bags, so 4 pounds makes 8 bags.* — Students see that 4 ÷ 1/2 = 8 and explain there are 2 halves in each whole pound.

Start

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **The detective can set up ___ markers because 10 ÷ 1/4 = ___.**
- **My answer is ___ because ___.**
  Mi respuesta es ___ porque ___.

Build

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ___.**
  Mi afirmación es ___.
- **My evidence is ___, so ___.**
  Mi evidencia es ___, entonces ___.

Explain

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- [ ] I answered the question.
- [ ] I used math evidence: numbers, an equation, or a model.
- [ ] I used at least two math words.

Writing Guide — Teacher Copy

- The answer to the math question is correct.
- The evidence is real lesson mathematics (numbers, equation, or model), not restated claim.
- The support level changed the language scaffolding, never the mathematical expectation.

Answer Key & Teacher Guide

1. **Notes 1:** Whole Number
2. **Notes 2:** Fraction
3. **Notes 3:** Reciprocal
4. **Notes 4:** Keep, Change, Flip
5. **Notes 5:** Quotient
6. **Watch for this mistake:** *A common mistake in Divide Whole Numbers by Fractions is doing 'Keep, Change' but skipping the 'Flip' — students multiply by the divisor as it is instead of by its reciprocal. For example, for 6 ÷ 3/4, a student might write 6 × 3/4 = 4 1/2 and stop there, but the divisor 3/4 must be flipped to 4/3 first: 6 × 4/3 = 24/3 = 8. Before you submit, check: did I flip the second fraction (the divisor) upside down before multiplying?*
7. **Try It 1:** B. 4 1/2 pouches (4 full, half a pouch left) — *3 ÷ 2/3 = 3 × 3/2 = 9/2 = 4 1/2. You can fill 4 full pouches with half a pouch of mix left over.*
8. **Try It 2:** C. 12 — *9 ÷ 3/4 = 9 × 4/3 = 36/3 = 12 pans.*